A Barvinok–Pataki Theorem for Archimedean Order Unit Spaces

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Abstract

We prove a generalisation of the Barvinok–Pataki rank bound to Archimedean order unit spaces equipped with a pointed cone. The classical result bounds the rank of extreme points of spectrahedra (feasible sets of semidefinite programs); our generalisation bounds the face dimension of feasible points under affine constraints in an arbitrary pointed cone, including the infinite-dimensional case. The face-dimension bound $d_F \leq m$ uses only pointedness of the cone and rank–nullity; finite-dimensionality of d_F in infinite dimensions follows from a result of Weis–Shirokov that requires only a real vector space and a convex set. When specialised to C*-algebras via the Alfsen–Shultz face–commutant correspondence, this recovers the operator-algebraic Barvinok–Pataki bound $\sum n_j^2 \leq m + 1$.

1 Preliminaries

Definition 1 (Ordered vector space with cone)**.** Let V be a real vector space. A *cone* is a subset $V^+ \subseteq V$ that is closed under addition and non-negative scalar multiplication. The pair (V, V^+) is an *ordered vector space* with partial order $x \leq y$ iff $y - x \in V^+$.

Definition 2 (Pointed cone)**.** A cone V^+ is *pointed* (or *proper*) if $V^+ \cap (-V^+) = \{0\}$, i.e. the only element v with both $v \geq 0$ and $-v \geq 0$ is $v = 0$.

Definition 3 (Order unit and Archimedean property)**.** An element $e \in V^+$ is an *order unit* if for every $v \in V$ there exists $n \in \mathbb{N}$ such that $ne - v \in V^+$ (i.e. $v \leq ne$). The cone V^+ is *Archimedean* if, whenever $ne + v \in V^+$ for all $n \in \mathbb{N}$, it follows that $v \in V^+$. The triple (V, V^+, e) is called an *Archimedean order unit space* (AOU space).

Definition 4 (Face of a cone)**.** Let V^+ be a cone in V and $x \in V^+$. The *face of V^+ generated by x* is

$$\mathrm{face}_{V^+}(x) \;=\; \left\{ w \in V^+ \;\middle|\; \exists\, \lambda > 0 \colon x - \lambda w \in V^+ \right\}.$$

Equivalently, $\mathrm{face}_{V^+}(x)$ consists of all elements of V^+ that can be "majorised" by a positive multiple of x. This is the smallest face of V^+ containing x in its relative interior.

Remark 5. In a pointed cone, $\mathrm{face}_{V^+}(x) = \{0\}$ if and only if $x = 0$. For $x \neq 0$, the face is at least one-dimensional (it contains the ray $\{\lambda x : \lambda \geq 0\}$). For the PSD cone PSD_n, the face generated by a rank-r matrix X is isomorphic to PSD_r, with $\dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$ (real symmetric case) or r^2 (complex Hermitian case).

2 The Classical Barvinok–Pataki Bound

We first recall the classical proof for PSD matrices, as it motivates each step of the generalisation.

Theorem 6 (Barvinok–Pataki, PSD case)**.** *Let $A_1, \dots, A_m \in \mathrm{Sym}_n(\mathbb{R})$ and $\beta_1, \dots, \beta_m \in \mathbb{R}$. Consider the spectrahedron*

$$C = \{\, X \in \mathrm{PSD}_n \mid \langle A_j, X\rangle = \beta_j,\ j = 1, \dots, m \,\}.$$

If $X \in C$ is an extreme point of C, then $\frac{r(r+1)}{2} \le m$ where $r = \mathrm{rank}(X)$.

Proof. Write $X = VBV^\top$ with $V \in \mathbb{R}^{n\times r}$, $B \in \mathrm{PSD}_r$ strictly positive definite. Define the compressed constraints $\hat{A}_j = V^\top A_j V \in \mathrm{Sym}_r$ and the linear map

$$\varphi\colon \mathrm{Sym}_r \to \mathbb{R}^m, \qquad [\varphi(D)]_j = \langle \hat{A}_j, D\rangle.$$

By rank–nullity, $\dim(\ker\varphi) \ge \frac{r(r+1)}{2} - m$. If $\ker\varphi \ne \{0\}$, choose $D \ne 0$ in $\ker\varphi$. Then $B(t) = B + tD$ satisfies all constraints for every t.

Since $B \succ 0$ and D is a non-zero symmetric matrix with at least one negative eigenvalue (replacing D by $-D$ if necessary), the map $t \mapsto \lambda_{\min}(B + tD)$ is continuous, positive at $t = 0$, and decreasing for $t > 0$. Let $t_0 > 0$ be the first value where $\lambda_{\min}(B + t_0 D) = 0$. Then $B(t_0) \succeq 0$ with $\mathrm{rank}(B(t_0)) < r$, and $VB(t_0)V^\top \in C$.

This contradicts X being an extreme point of C (since X is a non-trivial convex combination of $VB(t_0)V^\top$ and $VB(-t_0')V^\top$ for suitably chosen t_0'), unless $\ker\varphi = \{0\}$, which forces $\frac{r(r+1)}{2} \le m$. □

3 Generalisation to Pointed Cones

The key observation is that the proof above uses only three ingredients:

1. **Rank–nullity** for the constraint map restricted to the face;
2. **Pointedness** of the cone, to guarantee that a line through an interior point of a face must exit the cone;
3. **Finite-dimensionality** of the face, to apply rank–nullity with finite dimensions.

Theorem 7 (Barvinok–Pataki for pointed cones)**.** *Let V be a real vector space, $V^+ \subseteq V$ a pointed cone, and $\alpha_1, \dots, \alpha_m \in V^*$ linear functionals. Define*

$$C = \{\, x \in V^+ \mid \alpha_j(x) = b_j,\ j = 1, \dots, m \,\}$$

for given $b_1, \dots, b_m \in \mathbb{R}$. Let $x \in C$ and let $F = \mathrm{face}_{V^+}(x)$ be the face of V^+ generated by x. Assume that $d_F := \dim(\mathrm{span}(F))$ is finite. Define the restricted constraint map

$$\varphi_F\colon \mathrm{span}(F) \to \mathbb{R}^m, \qquad [\varphi_F(v)]_j = \alpha_j(v).$$

If x is an extreme point of C, then

$$\boxed{d_F \le m.}$$

Equivalently, $\dim(F) \le m - 1$ (where $\dim(F)$ is the dimension of F as a convex set, i.e. $\dim(\mathrm{aff}(F))$).

Proof. By rank–nullity applied to φ_F,

$$d_F \;=\; \dim(\ker\varphi_F) + \dim(\operatorname{im}\varphi_F) \;\le\; \dim(\ker\varphi_F) + m.$$

It suffices to show that $\ker\varphi_F = \{0\}$ when x is extreme in C. Suppose for contradiction that there exists $D \in \ker\varphi_F$, $D \neq 0$.

Step 1: The perturbation preserves feasibility. Since $D \in \operatorname{span}(F) \cap \ker\varphi_F$, we have $\alpha_j(D) = 0$ for all j. Therefore, for any $t \in \mathbb{R}$:

$$\alpha_j(x + tD) = \alpha_j(x) + t\,\alpha_j(D) = b_j.$$

So $x + tD$ satisfies all affine constraints whenever it lies in V^+.

Step 2: Pointedness forces exit from the cone. Since V^+ is pointed, we cannot have both $D \in V^+$ and $-D \in V^+$ (this would imply $D = 0$). Therefore, at least one of the half-lines $\{x + tD : t > 0\}$ or $\{x - tD : t > 0\}$ must eventually leave V^+.

More precisely: since x lies in the relative interior of F, for sufficiently small $|t|$ we have $x + tD \in F \subseteq V^+$ (as $D \in \operatorname{span}(F)$ and x is an interior point of F). But the line cannot remain in V^+ in both directions indefinitely, since this would imply D and $-D$ are both in V^+ (by rescaling), contradicting pointedness.

Step 3: Walking to a smaller face. Let

$$t^+ = \sup\{t > 0 : x + tD \in V^+\}, \qquad t^- = \sup\{t > 0 : x - tD \in V^+\}.$$

By Step 2, at least one of t^+, t^- is finite. Say $t^+ < \infty$ (the other case is symmetric).

If V^+ is closed (which holds in finite dimensions, or under the Archimedean assumption in the norm topology), then $x^* := x + t^+D \in V^+$, and x^* lies on the boundary of V^+ relative to $\operatorname{span}(F)$. This means x^* belongs to a *proper face* $F' \subsetneq F$, so $\dim(F') < \dim(F)$. Moreover, x^* satisfies all constraints, so $x^* \in C$.

Step 4: Contradiction with extremality. We have $x^* = x + t^+D \in C$ and (if $t^- > 0$, which holds since x is interior to F) also $x^{**} = x - \varepsilon D \in C$ for small $\varepsilon > 0$. Then

$$x = \frac{\varepsilon}{t^+ + \varepsilon}\, x^* + \frac{t^+}{t^+ + \varepsilon}\, x^{**},$$

exhibiting x as a non-trivial convex combination of two distinct elements of C. If x were an extreme point of C, this is a contradiction.

Therefore $\ker\varphi_F = \{0\}$, and we conclude $d_F \le m$. □

4 Application to C*-Algebras and AOU Spaces

Corollary 8 (Barvinok–Pataki for AOU spaces)**.** *Let (V, V^+, e) be an Archimedean order unit space with $\dim(V) < \infty$. Let $\alpha_1, \ldots, \alpha_m \in V^*$ and consider the state-like set*

$$C = \big\{\, x \in V^+ \;\big|\; \alpha_j(x) = b_j,\ j = 1, \ldots, m \,\big\}.$$

For any extreme point x of C, the face of V^+ generated by x satisfies $\dim(\operatorname{face}_{V^+}(x)) \le m - 1$.

Proof. The Archimedean property implies the cone V^+ is closed (in the order-unit norm) and pointed. In finite dimensions, every face is finite-dimensional. Apply Theorem 7. □

Corollary 9 (Barvinok–Pataki for C*-algebras)**.** *Let A be a unital C^*-algebra and $a_1, \ldots, a_m \in A_{\mathrm{sa}}$. Consider the affinely constrained state space*

$$C = \big\{ \varphi \in S(A) \;\big|\; \varphi(a_j) = \beta_j,\ j = 1, \ldots, m \big\},$$

where $S(A) = \{\varphi \in A^* : \varphi \geq 0,\ \varphi(\mathbf{1}) = 1\}$. If φ is an extreme point of C, then

$$\dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} \;\leq\; m + 1,$$

where π_φ is the GNS representation of φ. Equivalently, if $\pi_\varphi(A)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ (Artin–Wedderburn), then $\sum_j n_j^2 \leq m + 1$.

Proof. The state space $S(A)$ is a base for the cone A_+^* (positive functionals), cut out by the normalisation constraint $\varphi(\mathbf{1}) = 1$. The total number of affine constraints on A_+^* is $m + 1$ (the m given constraints plus normalisation).

By the Alfsen–Shultz face–commutant correspondence, the face of $S(A)$ generated by φ is affinely isomorphic to the state space $S(\pi_\varphi(A)')$ of the commutant. In particular,

$$\dim\big(\mathrm{face}_{S(A)}(\varphi)\big) = \dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} - 1.$$

Applying Theorem 7 with the $m+1$ constraints on A_+^* gives $\dim(\mathrm{span}(\mathrm{face}(\varphi))) \leq m+1$, hence $\dim_{\mathbb{R}}(\pi_\varphi(A)')_{\mathrm{sa}} \leq m + 1$. □

5 The Infinite-Dimensional Case and the Role of d_F

Theorem 7 assumes that $d_F = \dim(\mathrm{span}(F))$ is finite. In infinite dimensions this is a non-trivial hypothesis: the face of a cone generated by a point can be infinite-dimensional. We now show that this hypothesis is *automatically satisfied* whenever the number of constraints is finite, using a result of Weis–Shirokov that is purely convex-geometric.

Theorem 10 (Weis–Shirokov [4, Theorem 3.3])**.** *Let V be a real vector space, $K \subseteq V$ a convex set, and $f\colon K \to \mathbb{R} \cup \{+\infty\}$ a generalised affine map. Let C denote a sublevel set $\{x \in K : f(x) \leq \alpha\}$ or a level set $\{x \in K : f(x) = \alpha\}$, and let $x \in C$. If the face $F_C(x)$ of C generated by x has dimension $m \in \mathbb{N}_0$, then the face $F_K(x)$ of K generated by x has dimension m or $m + 1$.*

The multi-constraint version follows by induction (Corollary 5 of [4]): if C is the intersection of ℓ sublevel and/or level sets of generalised affine maps, and $F_C(x)$ has dimension m, then $\dim(F_K(x)) \in \{m, m + 1, \ldots, m + \ell\}$.

Corollary 11 (Finiteness of d_F)**.** *Let K be a convex set in a real vector space V, and let C be defined by ℓ generalised affine constraints on K. If x is an extreme point of C, then*

$$\dim\big(F_K(x)\big) \;\leq\; \ell.$$

In particular, the face of K generated by any extreme point of C is finite-dimensional, regardless of $\dim(V)$.

Proof. Since x is extreme in C, the face $F_C(x) = \{x\}$ has dimension 0. By the multi-constraint version of Theorem 10, $\dim(F_K(x)) \in \{0, 1, \ldots, \ell\}$. □

Remark 12 (No operator-algebraic assumptions needed)**.** Theorem 10 and Corollary 11 require only a real vector space V and a convex subset $K \subseteq V$. In particular:

- No cone structure, pointedness, or Archimedean property is needed to establish that d_F is finite.

- No C*-algebraic or operator-algebraic notions are needed.
- The operator-algebraic machinery (GNS representation, face–commutant correspondence, Artin–Wedderburn decomposition) is used only in a second step: to *interpret* the finite-dimensional face algebraically as a commutant and extract the multiplicity bound $\sum n_j^2 \leq m+1$.

This clarifies the logical structure of the infinite-dimensional case: Weis–Shirokov provides the purely geometric bound on d_F, and the Alfsen–Shultz correspondence translates it into algebra.

Combining Corollary 11 with Theorem 7, we obtain the following infinite-dimensional version of the main theorem that does not require an a priori finiteness assumption on d_F.

Corollary 13 (Barvinok–Pataki for infinite-dimensional pointed cones)**.** *Let V be a (possibly infinite-dimensional) real vector space, $V^+ \subseteq V$ a pointed closed cone, and $\alpha_1, \ldots, \alpha_m \in V^*$ linear functionals. Define $C = \{x \in V^+ : \alpha_j(x) = b_j\}$. If x is an extreme point of C, then*

$$d_F = \dim\big(\mathrm{span}(\mathrm{face}_{V^+}(x))\big) \leq m.$$

Proof. By Corollary 11 (with $K = V^+$ viewed as a convex set and $\ell = m$), d_F is finite. Theorem 7 then gives $d_F \leq m$. □

6 Special Cases Recovered

1. **Classical Barvinok–Pataki** ($A = M_n(\mathbb{C})$). The unique irreducible representation is the identity, so $\pi_\varphi(A)' \cong M_r(\mathbb{C})$ where $r = \mathrm{rank}(\rho_\varphi)$. Hence $r^2 \leq m+1$, or $r^2 \leq m$ when one of the constraints is the trace normalisation already built into the spectrahedron.
2. **Carathéodory's theorem** ($A = C(X)$, abelian). All irreducible representations are one-dimensional, so the commutant decomposes as $\bigoplus_{j=1}^k \mathbb{C}$. The bound gives $k \leq m+1$: extreme measures under m moment constraints are supported on at most $m+1$ points.
3. **Block-diagonal algebras** ($A = \bigoplus_i M_{d_i}(\mathbb{C})$). The bound $\sum_i r_i^2 \leq m+1$ simultaneously constrains ranks within blocks *and* the number of active blocks.
4. **Infinite-dimensional** ($A = B(\mathcal{H})$). For normal states (density operators) under m constraints (including possibly unbounded Hamiltonians, handled via the Weis–Shirokov framework): $r^2 \leq m+1$. In particular, $m \leq 2$ constraints imply all extreme points are pure states.

7 Physical Example: Hydrogen Atom in a Magnetic Field

We illustrate the theorem with a concrete physical system: the hydrogen atom in an external magnetic field, subject to angular momentum constraints. This example demonstrates how the Barvinok–Pataki bound determines when pure-state variational methods suffice and when mixed states are unavoidable.

7.1 Hilbert space and quantum numbers

The non-relativistic hydrogen atom has Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ (spatial wave function tensored with spin-$\frac{1}{2}$). The Coulomb Hamiltonian

$$H_0 = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0\, r}$$

has eigenvalues $E_n = -13.6\ \mathrm{eV}/n^2$ for $n = 1, 2, 3, \ldots$, with simultaneous eigenstates $|n, l, m_l, m_s\rangle$ labelled by:

Quantum number	Range	Observable
n (principal)	$1, 2, 3, \ldots$	H_0 (energy)
l (orbital ang. mom.)	$0, 1, \ldots, n-1$	$\mathbf{L}^2 = l(l+1)\hbar^2$
m_l (magnetic)	$-l, -l+1, \ldots, l$	$L_z = m_l\hbar$
m_s (spin)	$\pm\frac{1}{2}$	$S_z = m_s\hbar$

The degeneracy of the n-th energy shell is $g_n = 2n^2$ (including spin). The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with $J_z = L_z + S_z$ having eigenvalue $m_j\hbar$ where $m_j = m_l + m_s$.

7.2 Zeeman Hamiltonian

In a uniform magnetic field $\mathbf{B} = B\hat{z}$, the interaction Hamiltonian is

$$H_Z = \frac{\mu_B B}{\hbar}\big(L_z + g_s\, S_z\big),$$

where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton and $g_s \approx 2$ is the electron spin g-factor. On the standard basis, $H_Z|n, l, m_l, m_s\rangle = \mu_B B(m_l + 2m_s)|n, l, m_l, m_s\rangle$. The full Hamiltonian is $H = H_0 + H_Z$.

7.3 The constrained optimisation problem

We seek the state of minimum energy subject to a constraint on the total angular momentum projection:

$$\min_{\rho}\ \mathrm{Tr}(H\rho) \qquad \text{subject to} \qquad \rho \geq 0,\ \ \mathrm{Tr}(\rho) = 1,\ \ \mathrm{Tr}(J_z\,\rho) = M\hbar. \tag{1}$$

This is a convex optimisation (linear objective over a convex constraint set). The minimum is attained at an extreme point of the feasible set

$$C = \big\{\rho \in \mathfrak{S}(\mathcal{H}) : \mathrm{Tr}(J_z\,\rho) = M\hbar\big\},$$

where $\mathfrak{S}(\mathcal{H})$ is the set of density operators on $\mathcal{H}$.

7.4 Applying the Barvinok–Pataki bound

Full Hilbert space ($A = B(\mathcal{H})$, infinite-dimensional). The constraint $\mathrm{Tr}(J_z\,\rho) = M\hbar$ constitutes $m = 1$ affine constraint on the state space $\mathfrak{S}(\mathcal{H})$. By Corollary 9 (or Corollary 13 applied to the positive cone in the trace-class operators),

$$r^2 \ \leq\ m + 1 = 2, \qquad \text{hence} \qquad r = 1.$$

The energy minimiser is a pure state.

More generally, with m constraints:

m	Constraints	Rank bound	Conclusion
1	$\mathrm{Tr}(J_z\,\rho) = M\hbar$	$r = 1$	Pure state
2	$+\ \mathrm{Tr}(\mathbf{J}^2\,\rho) = j(j+1)\hbar^2$	$r = 1$	Pure state
3	$+\ \mathrm{Tr}(\mathbf{L}^2\,\rho) = l(l+1)\hbar^2$	$r \leq 2$	Rank ≤ 2
4	$+\ \mathrm{Tr}(H_0\,\rho) = E$	$r \leq 2$	Rank ≤ 2

The transition from pure to potentially mixed occurs at $m = 3$: this is a direct consequence of the face dimension gap for $B(\mathcal{H})$, which has $\{1, 2\} \subseteq G(B(\mathcal{H}))$ (the face dimension spectrum is $\{0, 3, 8, 15, \ldots\}$).

Restriction to the $n = 2$ shell. For a concrete finite-dimensional calculation, restrict to the $n = 2$ shell. This has $g_2 = 2 \cdot 4 = 8$ states, so $\mathcal{H}_{n=2} \cong \mathbb{C}^8$ and $A = M_8(\mathbb{C})$:

l	m_l	m_s	$m_j = m_l + m_s$	$m_l + 2m_s$	**Subshell**
0	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2s$
0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2s$
1	-1	$+\frac{1}{2}$	$-\frac{1}{2}$	0	$2p$
1	-1	$-\frac{1}{2}$	$-\frac{3}{2}$	-2	$2p$
1	0	$+\frac{1}{2}$	$+\frac{1}{2}$	1	$2p$
1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	-1	$2p$
1	1	$+\frac{1}{2}$	$+\frac{3}{2}$	2	$2p$
1	1	$-\frac{1}{2}$	$+\frac{1}{2}$	0	$2p$

The Coulomb Hamiltonian is $H_0 = E_2\, I_8$ (degenerate on the shell). Including spin–orbit coupling (fine structure), $H_{\text{fs}} = \xi_{21}\, \mathbf{L} \cdot \mathbf{S}$ splits the shell into multiplets: $2s_{1/2}$ ($l = 0$, $j = \frac{1}{2}$, 2 states), $2p_{1/2}$ ($l = 1$, $j = \frac{1}{2}$, 2 states), and $2p_{3/2}$ ($l = 1$, $j = \frac{3}{2}$, 4 states). The Zeeman term $H_Z = \mu_B B(m_l + 2m_s)$ is diagonal in the $|l, m_l, m_s\rangle$ basis.

The optimisation (1) restricted to this shell is an SDP on $M_8(\mathbb{C})$. Normalisation $\mathrm{Tr}(\rho) = 1$ is built into the state space. The Barvinok–Pataki bound gives:

m	**Constraints (on the $n = 2$ shell)**	$r^2 \le m + 1$	**Naïve bound**
1	$\mathrm{Tr}(J_z\, \rho) = M\hbar$	$r = 1$	$r \le 8$
2	$+\ \mathrm{Tr}(\mathbf{J}^2\, \rho) = j(j+1)\hbar^2$	$r = 1$	$r \le 8$
3	$+\ \mathrm{Tr}(\mathbf{L}^2\, \rho) = l(l+1)\hbar^2$	$r \le 2$	$r \le 8$
5	$+\ \mathrm{Tr}(S_z\, \rho) = s\hbar,\ \mathrm{Tr}(L_z\, \rho) = \mu\hbar$	$r \le 2$	$r \le 8$

7.5 Physical interpretation

1. **Automatic purity ($m \le 2$).** Fixing the total angular momentum projection M (and optionally j) is enough to guarantee that the energy minimiser is a pure state. This means pure-state variational methods (e.g. Rayleigh–Ritz, DMRG for the ground state) are *exact* for this problem — there is no advantage to working with density matrices.

2. **Low-rank mixed states ($m = 3$).** Adding a third constraint (e.g. fixing l) opens the door to rank-2 optimisers: the energy minimiser may be a *statistical mixture* of two pure states. But the rank is at most 2, not 8; thus only a two-dimensional search is needed, rather than optimisation over all 8×8 density matrices.

   This bound is **tight**: when the constraint values lie in the interior of the joint numerical range — i.e. values not achievable by any single pure state — the optimizer is genuinely mixed. For example, on the $n = 2$ shell with $m = 3$ constraints $\langle J_z \rangle = 0$, $\langle \mathbf{J}^2 \rangle = 2$, $\langle \mathbf{L}^2 \rangle = 1$ (values strictly between the eigenvalues of $\mathbf{J}^2$), the SDP optimizer has $\mathrm{rank}(\rho^*) = 2$ with eigenvalues $(\lambda_1, \lambda_2) \approx (0.569, 0.431)$. See Table 1 and Figure 1.

3. **Practical consequence.** For a numerical SDP solving (1), interior-point methods converge to the analytic centre (maximal rank), but the theorem guarantees that facial reduction or rank minimisation will find a solution of rank $\le \lfloor \sqrt{m+1} \rfloor$. This dramatically reduces the dimension of the search space: on the $n = 3$ shell ($g_3 = 18$, $\rho \in M_{18}(\mathbb{C})$), even $m = 10$ constraints yield $r \le 3$ rather than $r \le 18$.

Table 1: Genuine rank-2 optimizers on hydrogen shells with $m = 3$ constraints (BP bound $r \leq 2$). Constraint values are chosen in the interior of the joint numerical range, where no pure state is feasible.

n	dim	$\langle J_z \rangle$	$\langle \mathbf{J}^2 \rangle$	$\langle \mathbf{L}^2 \rangle$	rank	λ_1	λ_2
2	8	0	2.0	1.0	**2**	0.569	0.431
2	8	0	1.5	0.5	**2**	0.754	0.246
3	18	0	4.0	3.0	**2**	0.675	0.325

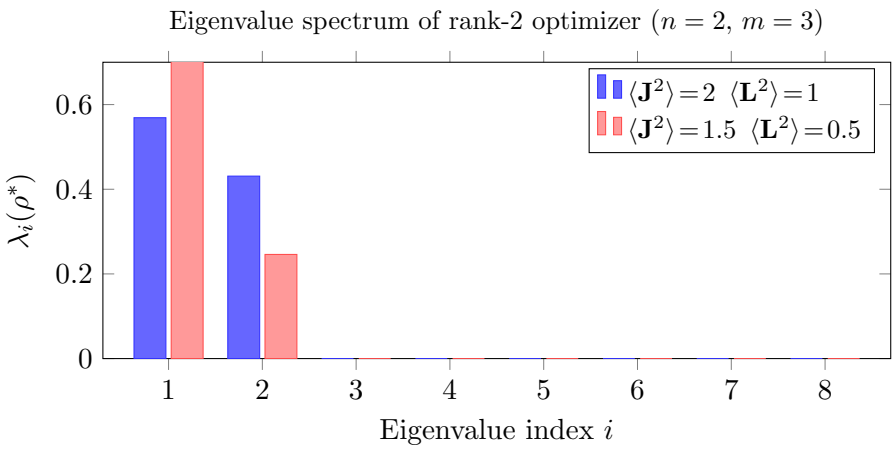

Figure 1: Eigenvalue spectra of two genuine rank-2 optimizers on the $n = 2$ shell (dim $= 8$) with $m = 3$ constraints and $\langle J_z \rangle = 0$. Exactly two eigenvalues are non-zero, saturating the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor = 2$. The constraint values lie in the interior of the joint numerical range and cannot be achieved by any pure state.

7.6 Numerical verification

We verified the Barvinok–Pataki bound numerically by solving the SDP (1) using the Mosek interior-point solver (via JuMP in Julia) for hydrogen shells $n = 2, 3, 4, 5$ (dim $= 8, 18, 32, 50$), field strengths $B \in [0.05, 5]$, and $m = 0, \ldots, 5$ angular momentum constraints. A two-phase approach was used: Phase 1 finds the optimal energy E^*; Phase 2 adds a small random Hermitian perturbation to the objective while pinning $\mathrm{Tr}(H\rho) = E^*$, generically selecting a unique extreme point of the optimal face.

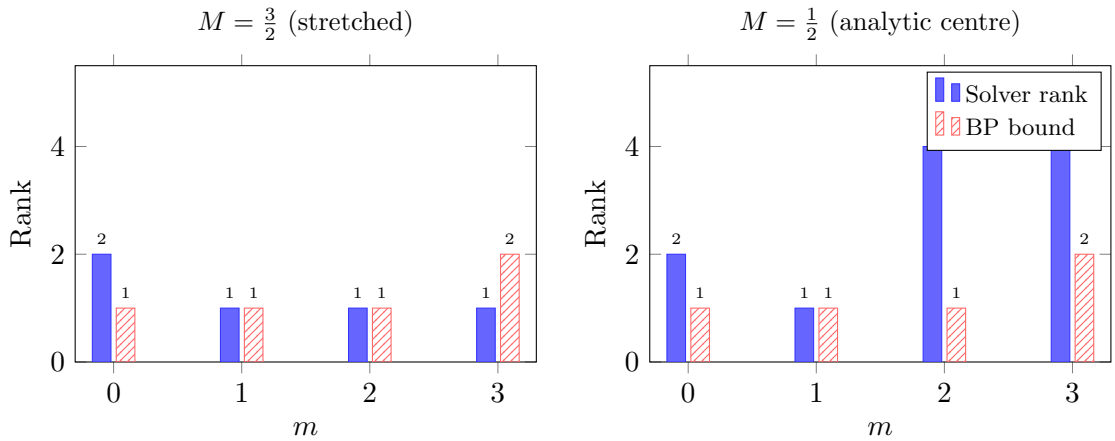

Figure 2: Optimizer rank on the $n = 2$ shell. **Left:** $M = \frac{3}{2}$ (maximally stretched); the extreme-point optimizer satisfies the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor$. **Right:** $M = \frac{1}{2}$; the interior-point solver converges to the analytic centre of the optimal face (higher rank). Facial reduction would recover an extreme point satisfying BP.

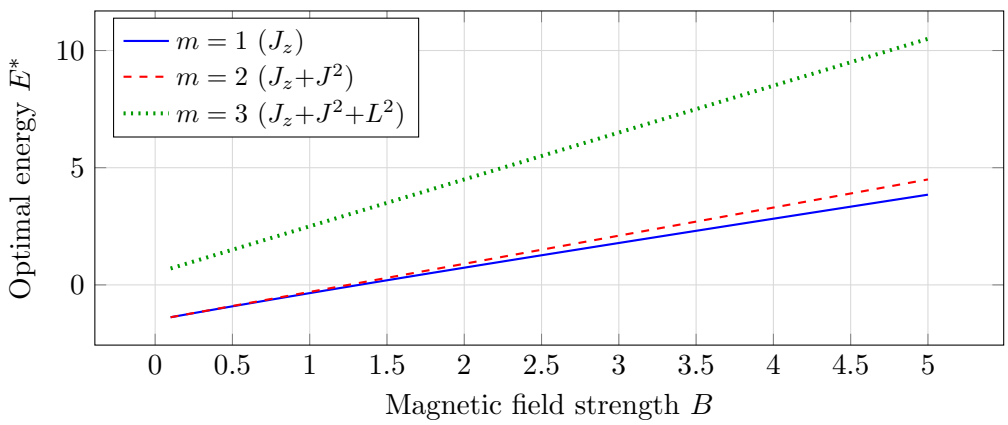

Figure 3: Optimal energy vs. field strength on the $n = 3$ shell (dim $= 18$, $M = \frac{3}{2}$). Adding constraints raises the minimum energy. All optimizers are pure ($r = 1$), consistent with BP.

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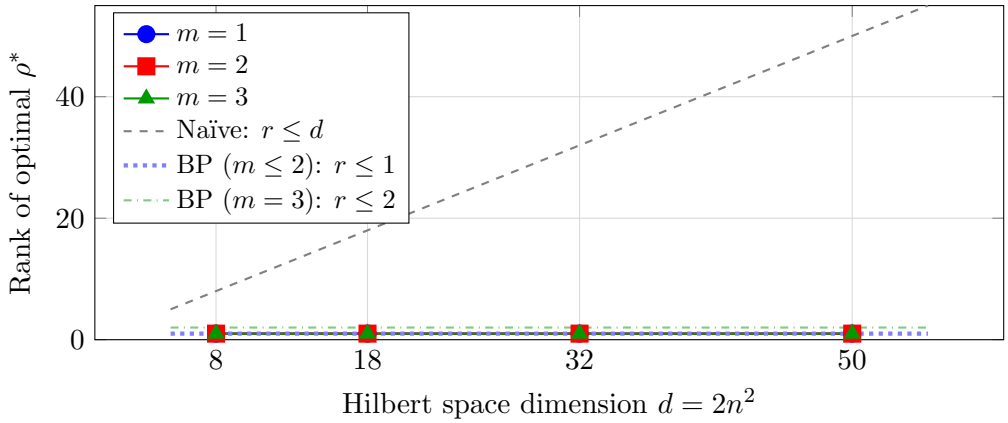

Figure 4: Optimizer rank vs. Hilbert space dimension ($B = 1$, $M = \frac{3}{2}$). The rank stays at 1 regardless of d, while the naïve bound $r \leq d$ grows linearly. For $n = 5$ ($d = 50$), BP reduces the rank bound from 50 to at most 2.

quantity d_F can be controlled in infinite dimensions via Theorem 3.3 of Weis–Shirokov [4], which requires only a real vector space and a convex set, without operator-algebraic assumptions. The recording was transcribed and typeset with the assistance of Claude (Anthropic).

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